

MIKHAIL SHYTSKO

mikhail.shytsko@gmail.com • [Personal page](#) • [LinkedIn](#) • +421950310445

SUMMARY

Software & AI engineer who ships production AI systems - agentic LLM applications, MCP-based agents, and the backends behind them. Experience ranges from LLM tooling and AI agents inside a regulated bank to founding an AI-native data-generation product, with a focus on making non-deterministic LLM systems measurable and reliable. Strong Spring/JVM background, including high-load banking microservices serving 2M+ customers, which is now applied to building AI/LLM systems in Kotlin and Python.

WORK EXPERIENCE

Solutions & AI Engineer, Slovenska Sporitelna, a.s.

Jul 2025 - Today

- Driving the design and build of an **MCP-based AI assistant** that summarizes and cross-references corporate-lending data spread across several internal banking systems - replacing the manual, multi-system lookup that slows credit-application review.
- Architected the AI's data access as a **least-privilege MCP tool layer** - fully auditable by design, with strict data minimization and a controlled disable path, so the assistant runs safely inside a regulated banking environment without altering the core approval workflow.
- Built **change-detection** into the assistant - surfacing and explaining shifts across products, limits, collateral, and client data, so reviewers catch material changes they'd otherwise miss across systems.
- Delivered a **six-figure cost reduction** by replacing a legacy workflow with an in-house **Kafka** pipeline - production-ready, end-to-end from a one-page spec.
- Cut response times **85%** for two high-load applications serving **2M+ customers** by taking sole ownership of inherited services and driving targeted performance optimizations.

Worked with: Kotlin, Spring Ecosystem, MCP, Azure OpenAI, LLMs, Apache Kafka

Founding AI Engineer, Seedfast

Oct 2025 - Today

- Built an AI system that generates **realistic, referentially-correct data** for any database, sparing AI, QA, and dev teams **days of manual data-engineering** per dataset in a single run.
- Re-architected the generation core into an **agentic code-execution model** - an LLM that writes and runs its own data-generation code - with independent output verification that eliminated a class of silent data-integrity failures.
- Engineered a **GEPA-based self-optimizing eval harness** (deterministic checks + LLM-as-judge) that more than doubled the system's combined success / quality / speed score on held-out evaluation - making a non-deterministic LLM pipeline measurable and regression-gated.
- Raised generated-data quality **from 28% to 73%** on a multi-dimensional benchmark (language fidelity, domain relevance, cross-table coherence) by fixing how user requirements propagate into generation; added locale-aware output in any language.
- Shipped the product's developer-facing **Go CLI** and a **Model Context Protocol (MCP) server** - letting both engineers and AI agents run product in production, CI/CD, and agentic workflows.

Worked with: Python, Go, LangGraph, GEPA, MCP, OpenAI API, LangSmith, PostgreSQL, AWS

Software Engineer, The Designery s.r.o.

Apr 2025 - Jul 2025

- Built a secure, EU GDPR-compliant backend from scratch on **Spring WebFlux + GCP** for a mental-health startup, encrypting sensitive patient data in real time and automating a CI/CD pipeline that cut deploys from 12 to 4 min.

Worked with: Java, Spring WebFlux, GCP (Firestore, Firebase Auth, Cloud Functions), GitLab

Software Engineer, VitSBS

May 2024 - Oct 2024

- Automated equipment certification, cutting document-certification time **~50%**, and built **GraalPy** trend analytics that accelerated product decisions **~25%**.

Worked with: Java, Spring, GraalVM/GraalPy, PostgreSQL, MongoDB, Docker

Backend Developer, ArenaPizza

Oct 2023 - Mar 2024

- Shipped a secure **Spring Boot** REST API (OAuth2/JWT) and cut shift-approval time **from 5 days to 2** through schedule optimization.

Worked with: Java, Spring Boot, OAuth2/JWT, PostgreSQL, Redis

EDUCATION AND ADDITIONAL INFORMATION

Bachelor of Computer Science - Technical University of Kosice

Achievements:

- 1st place for **Seedfast** at [CapTTict's DDAccelerator Finals](#) - the only Slovak startup competing across 9 countries.
- Languages: English, Slovak, Russian, Belarussian
- Interests: playing music & table tennis